Retail Sales Analysis Project

Excel • MySQL • Power BI

Project Overview

The objective of this project was to analyze retail sales data and develop an interactive business intelligence dashboard using Excel, MySQL, and Power BI. The project follows a complete data analytics workflow beginning with data cleaning, continuing through SQL-based exploratory analysis, and concluding with an interactive dashboard that provides meaningful business insights.

The dataset contains more than 10,000 retail transactions and includes information related to customers, products, sales, profit, discounts, geographic regions, and order dates. By combining multiple analytical tools, this project demonstrates the ability to clean raw data, perform business analysis with SQL, and communicate findings through data visualization.

Data Preparation

The project began by importing the raw Superstore dataset into Microsoft Excel. Several data preparation tasks were completed to improve data quality before analysis. Duplicate records were removed, formatting inconsistencies were corrected, column widths were adjusted for readability, and a Profit Margin column was added to provide an additional business metric. The cleaned dataset was then exported as a CSV file for use in MySQL and Power BI.

SQL Analysis

After cleaning the dataset, it was imported into MySQL where several SQL queries were written to answer key business questions. Aggregate functions, filtering, sorting, grouping, and ordering techniques were used throughout the analysis.

Examples of business questions included:

- What are the company's total sales and total profit?
- Which geographic regions generate the highest revenue?
- Which product categories produce the greatest profit?
- Which customer segment contributes the largest share of sales?

Dashboard Development

Using Power BI, an interactive executive dashboard was developed to summarize business performance. KPI cards were created to display Total Sales, Total Profit, Total Orders, and Average Discount. Additional visualizations were designed to highlight regional sales performance, category profitability, customer segmentation, monthly sales trends, and the highest-selling products.

Interactive slicers were added to allow users to filter the dashboard by Region, Category, Segment, and other fields, providing a more dynamic user experience and allowing stakeholders to quickly explore different areas of the business. The dashboard was designed to present information in a clear and organized format suitable for executive decision-making.

Key Findings

Several meaningful business insights were identified throughout the analysis.

The **West** region generated the highest sales revenue, while the **East** region followed closely behind. This indicates that these markets contribute a significant portion of the company's overall revenue and may represent opportunities for continued investment and customer growth.

Customer segmentation revealed that Consumer customers account for the largest percentage of total sales, making this group the company's primary revenue source. Corporate and Home Office customers contribute a smaller share but represent potential opportunities for targeted marketing campaigns.

Product-level analysis demonstrated that a relatively small number of products generate a disproportionately large percentage of total revenue. This suggests that maintaining inventory availability and promotional efforts for these products could positively impact future sales performance.

Monthly sales trends illustrate fluctuations in revenue over time, providing management with valuable information for seasonal planning, inventory management, and forecasting future demand.

Finally, the average discount metric highlights the relationship between promotional pricing and profitability, emphasizing the importance of balancing customer incentives with sustainable profit margins.

Business Recommendations

Based on the analysis, several recommendations can be made.

The organization should continue investing in high-performing regions while investigating factors limiting growth in lower-performing markets. Inventory planning should prioritize the highest-selling products to minimize stock shortages. Marketing efforts may be expanded toward Corporate and Home Office customer segments to diversify revenue sources.

Additionally, discount strategies should be reviewed regularly to ensure promotional campaigns continue supporting both customer acquisition and long-term profitability.

Technical Skills Demonstrated

This project demonstrates practical experience with multiple data analytics tools and techniques, including:

- Microsoft Excel Data Cleaning
- CSV Data Preparation
- SQL Query Development
- Data Aggregation and Business Analysis
- MySQL Database Management
- Power BI Dashboard Development
- KPI Design
- Interactive Data Visualization
- Business Intelligence Reporting
- Data Storytelling

Conclusion

This project demonstrates an end-to-end data analytics workflow beginning with raw transactional data and ending with an interactive business dashboard. By combining Excel, SQL, and Power BI, the analysis transforms large volumes of retail data into meaningful insights that support business decision-making.

Completing this project strengthened practical skills in data preparation, database querying, dashboard development, and business communication while reinforcing the importance of presenting analytical findings in a clear and actionable manner. The project serves as a portfolio example that reflects the complete lifecycle of a business intelligence solution and demonstrates the ability to convert raw data into valuable business insights.